CM52065 – Natural Language Processing

Support Ticket classification

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<u>Set</u>: 25/02/2026, 12:15 **<u>Due</u>: 20/03/2026, 20:00**
<u>Percentage of overall unit mark</u>: 30%
<u>Submission Location</u>: Moodle **<u>Submission Components</u>: Report & Code** **<u>Submission Format</u>: 2 x Jupyter Notebook (.ipynb) and 1 x Report (.pdf)**
<u>Anonymous Marking</u>: Y

1 Overview

This coursework assesses two of the three module Learning Outcomes:

1. Demonstrate knowledge of the fundamental principles of natural language processing.
2. Demonstrate the ability to implement basic algorithms, and to effectively use and evaluate existing implementations of more complex algorithms.

1.1 The Task

Providing IT support requires a range of expertise to handle an ever-changing stream of queries and questions from users. You are tasked with developing a model capable of classifying user queries into one of several categories. To do this, you will need to:

1. Analyse the domain of the user queries.
2. Implement two different pre-processing pipelines which convert the user queries from their raw state to a representation suitable for classification.
    a. The pipelines must differ in at least one way, for example, the use of Stemming vs. Lemmatisation.
3. Select, train and evaluate one classification model for each of your pre-processing pipelines to classify the processed user queries.
4. Document your approach and findings in a short-form report.

To help you achieve this goal a labelled dataset of queries has been provided and is described in Section 3.

2 Deliverable

There are two deliverables to this coursework:

<table>
<tr><th>Deliverable</th><th>Format</th><th>Content</th></tr>
<tr><td>Code</td><td>2 x .ipynb</td><td>Two Jupyter Notebooks each containing the code necessary to reproduce one of the pipelines. Each notebook must contain:
Code to load and perform text preprocessing on the data provided.
Code to reproduce your models.
Code to reproduce your evaluation and results.</td></tr>
<tr><td>Report</td><td>PDF</td><td>A report containing details of your model development, the report must contain the following sections:
Introduction: A high-level description of the problem.
Data Handling: Overviews of the two pre-processing pipelines, highlighting their differences.
Modelling: Description of the classification approach taken.
Evaluation: Detailed quantitative and qualitative evaluation of the classification models.
Conclusion: Reflection on the results, approach, data and implementation.
The report must be submitted in PDF format using a reasonable font (Arial, Calibri etc.) at size 11 minimum. The maximum page count is 5 excluding references.</td></tr>
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2.1 Mark Scheme

Your report and code will be assessed using the contributions below:

Component	Criteria	Percentage of overall mark
Report (80%)	Introduction	10%
	Data Handling	25%
	Modelling	10%
	Evaluation	25%
	Conclusions	10%
Jupyter Notebooks (20%)	Reproducibility & Presentation	20%

Table 2. Mark Allocation.

Outlines for what would be expected of a typical submission in each grade classification are as follows:

- Distinction (>70%): Exceptional implementation which goes above and beyond the standard practices of pre-processing to provide a unique approach specific to the use-case presented. The classification models are trained and then evaluated using techniques beyond the standard classification metrics with detailed discussion on the qualitative and quantitative results. The report not only details the approach and results but provides a detailed reflection not limited to the process implemented.
- Merit (60-69%): Considered implementations of two pre-processing pipelines are provided with good use of external libraries and functions to overcome the challenges posed. Classification models are trained and have undergone thorough extrinsic evaluation using both quantitative and qualitative methods. The report not only details the process but provides insight, reflections and justifications tied to the use-case provided.
- Pass (50-59%): Two pre-processing pipelines are implemented with key differences in their approach and classifiers are trained and evaluated based on the output of the pre-processing pipeline. The report details the process used in all stages of the experiment, providing superficial justification and reflections.
- Implementation of a text preprocessing pipeline and evaluation of two classifiers. The report details the primary components of the methodology with some justifications and reflections.
- Fail (<50%): Inadequate implementation of text preprocessing pipeline, classifier and/or evaluation methods. The report contains minimal details, lacks justification and/or provides superficial and limited reflection.

3 The Dataset

You have been provided a dataset in the form of a CSV file containing two columns. The first column contains the text of the support request, the second column contains the manually assigned class for the request. In total there are 308 labelled samples spread across 14 different support ticket types. The table below provides an overview of the number of samples for each class.

Support Class	**No. Samples**
shipping_delay	29
order_tracking	22
update_email	23
account_recovery	21
billing_history	21
data_privacy	22
payment_issue	33
reset_password	20
subscription_management	23
delete_account	23
refund_request	21
notification_settings	24
login_issue	16
two_factor_auth	10

Table 1. Number of samples per support ticket class.

4 Feedback

Formative feedback will be available in the labs.

You will receive **summative feedback** on your work within 3 semester weeks of the submission deadline. The feedback will discuss your performance based on the criteria for marking, including what you did well and how specific components/sections could have been improved.

5 Academic Integrity

Your work will be checked to ensure that you have not plagiarised. For more information about the plagiarism policy at the University see: https://library.bath.ac.uk/referencing/plagiarism

Remember that published work that you refer to in your report should be clearly referenced in your text and listed in a bibliography section given at the bottom of your report. For more information see, https://library.bath.ac.uk/referencing/new-to-referencing

This coursework is classified as Type A with regards to the use of Generative AI, hence no generative AI should be used.

6 FAQ

6.1 Is the quality of my models marked?

In short, no. I am more interested in whether you can adequately build a text pipeline, evaluate your models' performance and reflect on the results. There are no direct marks available for the quality of the model.

6.2 Why is the page count so small?

Whether writing a technical report for a client or an academic conference paper concise writing is key. You may use the marking weights as a rough guide as to how long a given section may be; for example, 1 page for data handling, ½ a page for modelling etc.

6.3 Do we have to write the pipeline, evaluation methods or models from first-principles?

Definitely not, writing methods from scratch could hinder the reproducibility of your results.